

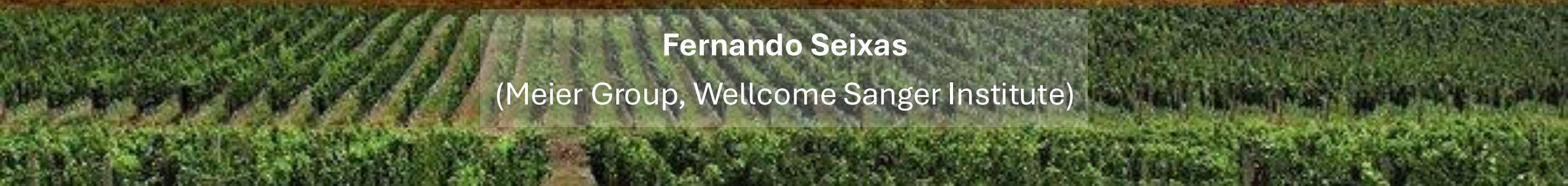
Genome Scans

2025 Biodiversity Genomics (Mendoza, Argentina)



Fernando Seixas

(Meier Group, Wellcome Sanger Institute)



Genome Scans

- What are genomes scans?

Analyses comparing genome-wide patterns of genetic variation across many loci and across the genome

- What can they be used for?

Selection: regions under natural selection, sources of selection.

Speciation: barriers to gene flow

Introgression: heterogeneity of introgression, adaptive introgression

- How are they applied?

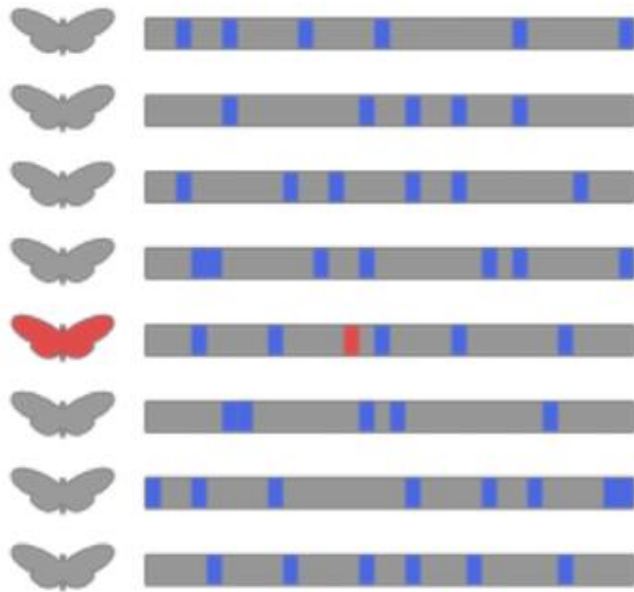
SNPs

Markers (e.g., when using reduced representation approaches – e.g RADseq, RNAseq)

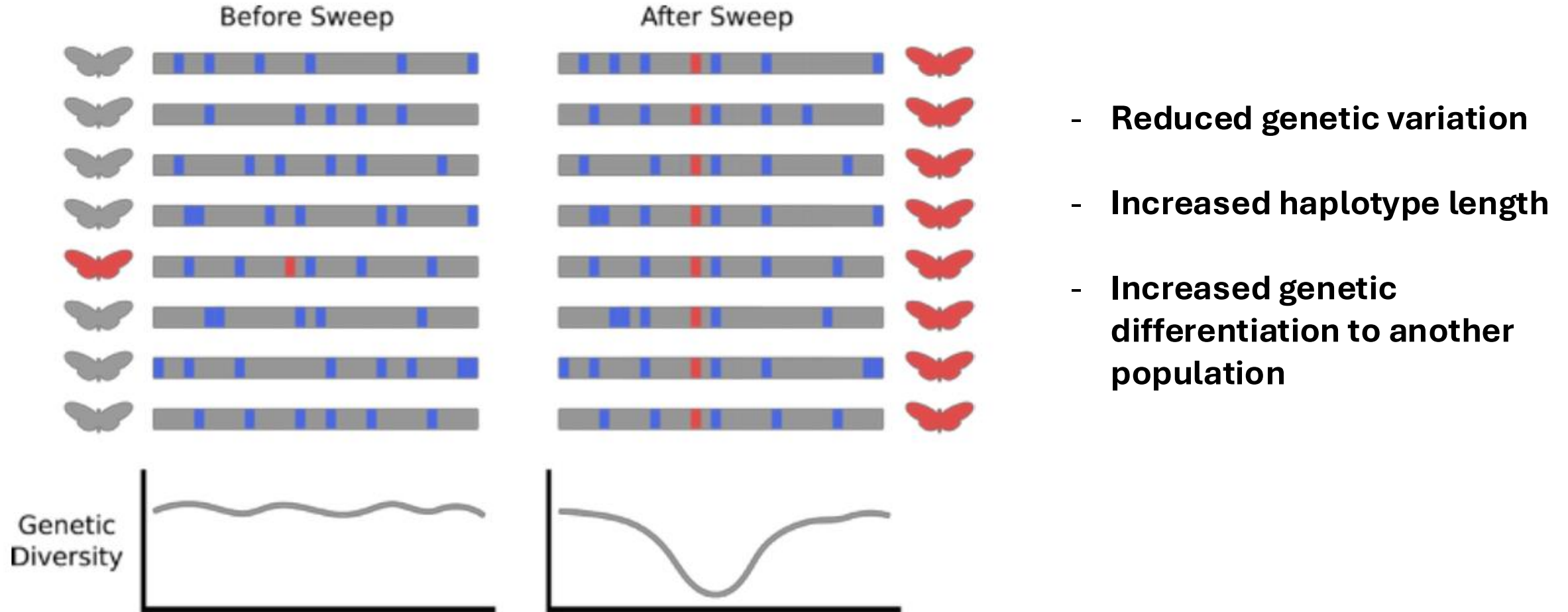
Sliding Window (either fixed physical windows or SNP windows)

Selective sweep signatures

Before Sweep



Selective sweep signatures



Many statistics

To detect barriers to gene flow or under selection

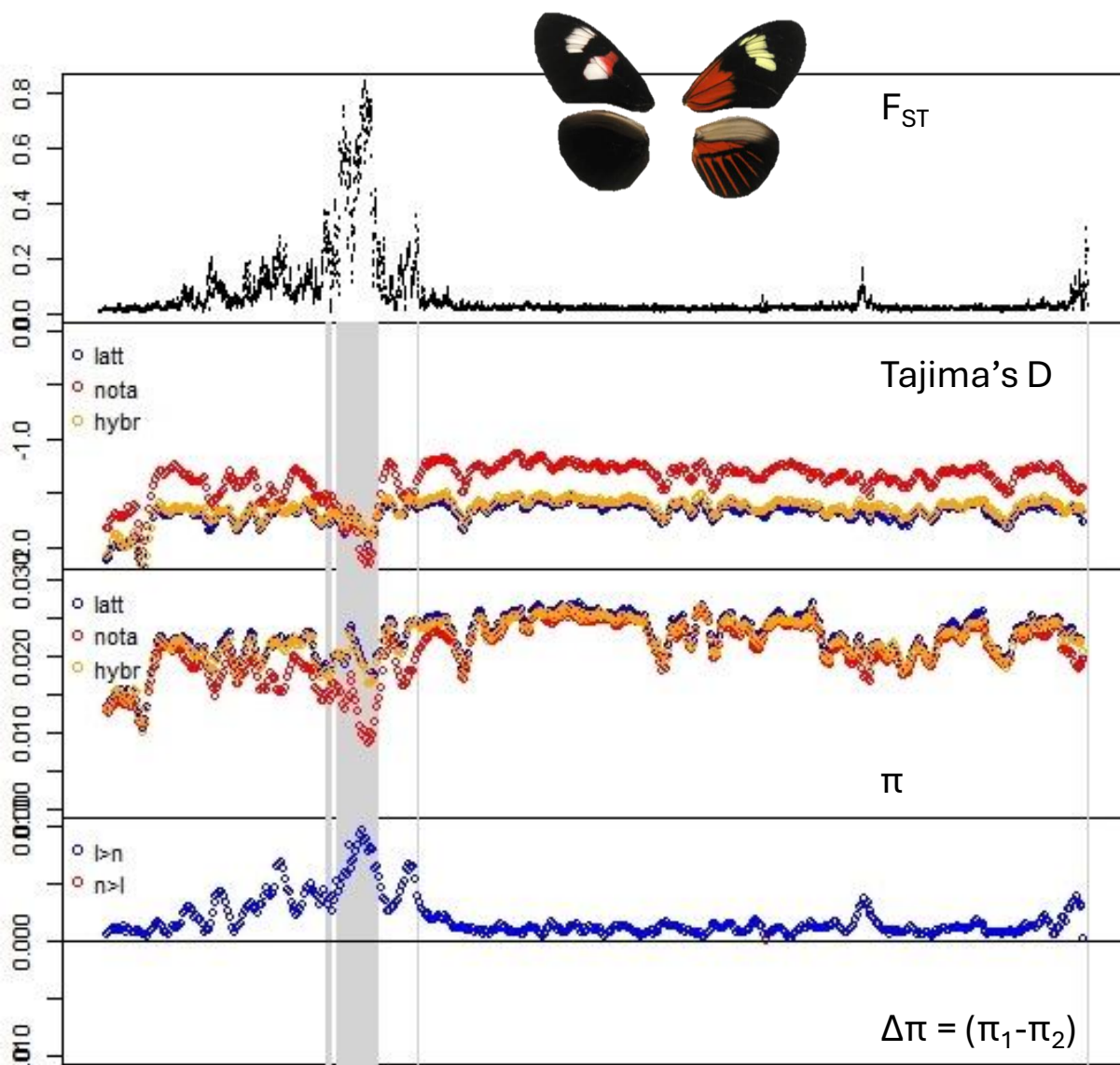
- π and $\Delta\pi$
- F_{st} , PBS
- d_{xy}
- Tajima's D
- Haplotype homozygosity tests (iHS, XP-EHH, H12) ^{*requires phasing}

see e.g. Wolf & Ellegren (2017) Nature Reviews Genetics

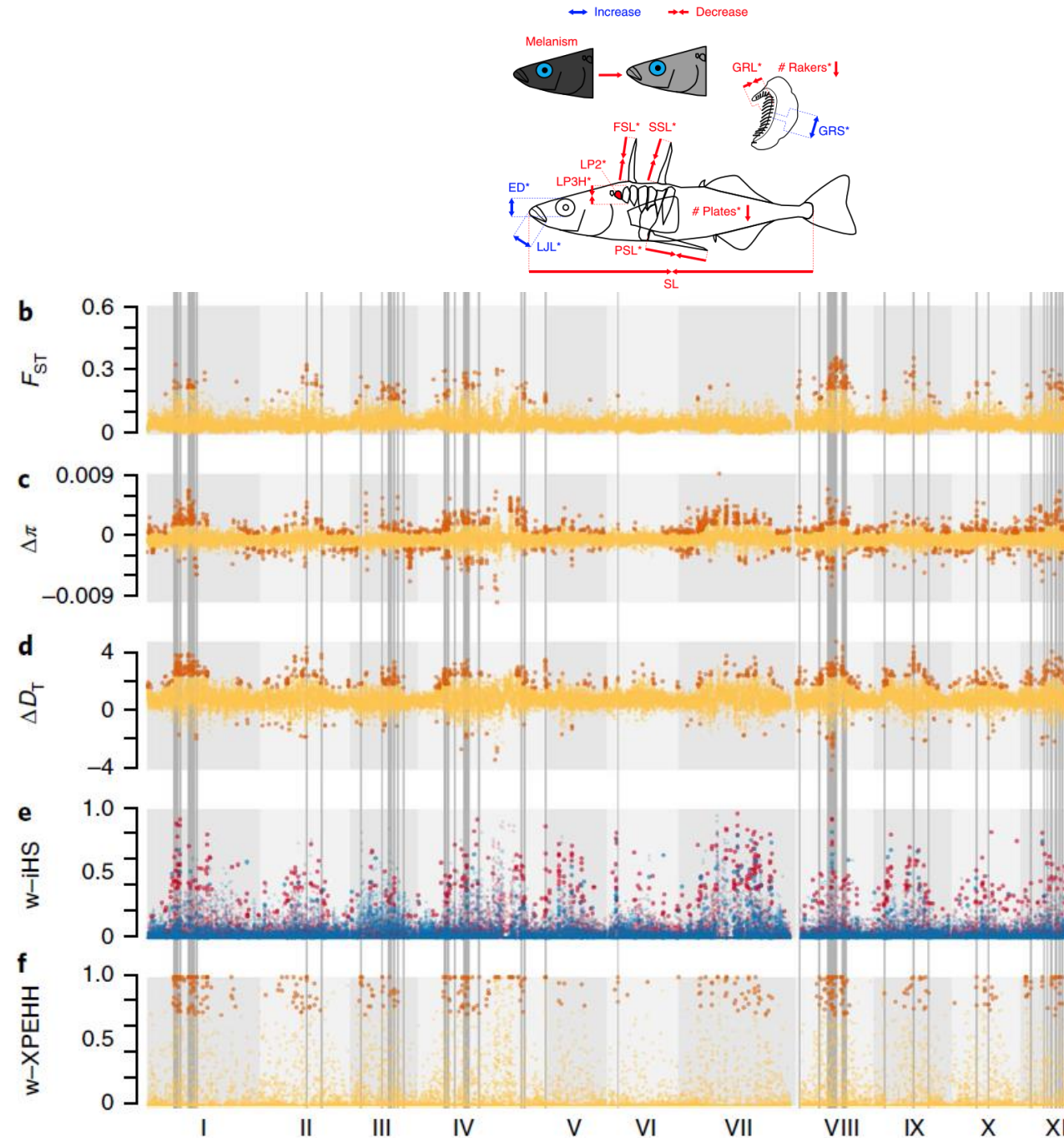
To detect regions of introgression:

- D-statistic, f_d (phylogenetic patterns)
- G_{min} (measure of divergence)
- TWISST (topology)

Combining different approaches

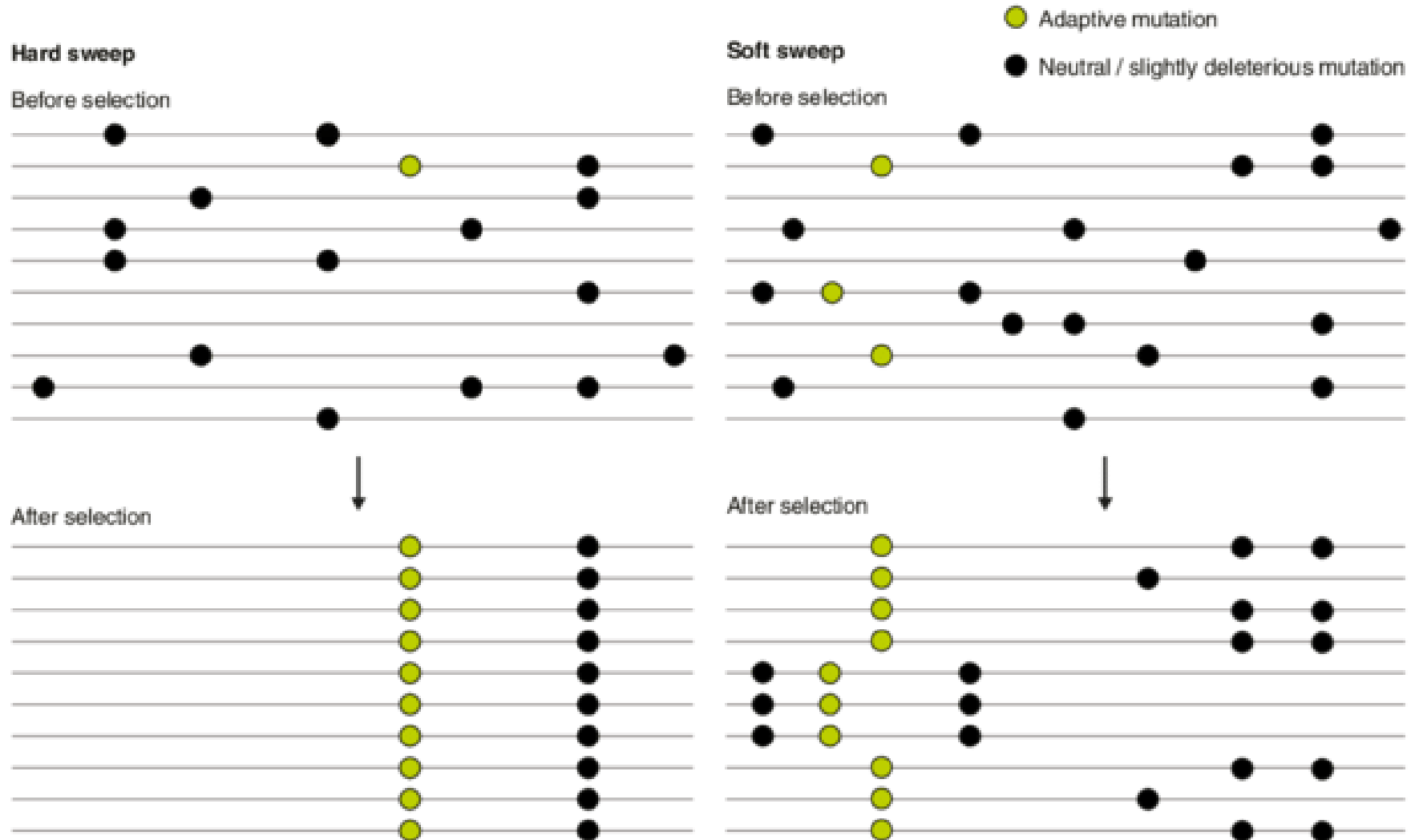


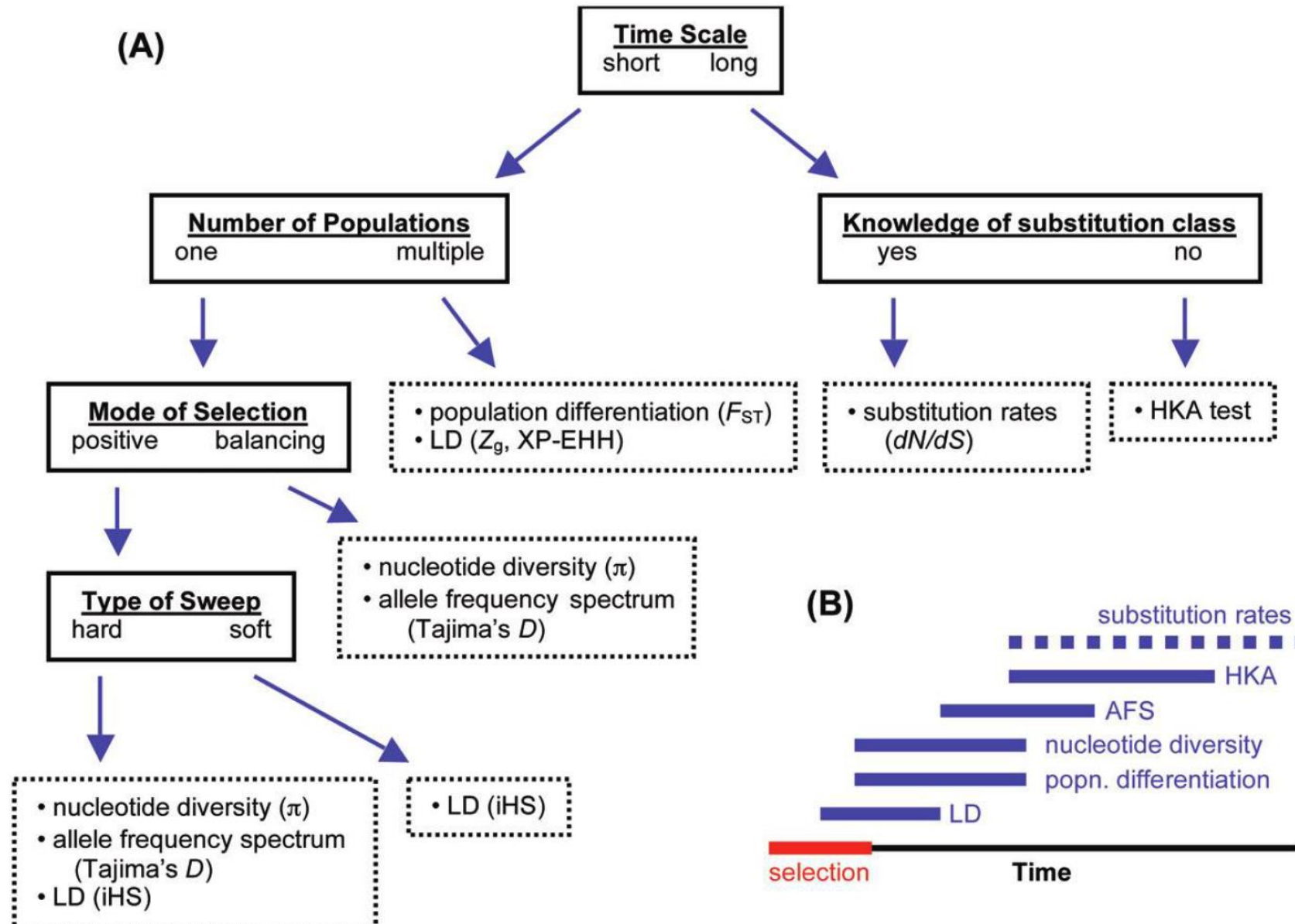
Meier et al., 2021, PNAS



Marques et al., 2018, Nature Ecol Evol

Hard sweep and soft sweep





Genome Scans and Speciation

Isolation view of Speciation

Dobzhansky (1937); Mayr (1942)

- Genome as the unit of speciation
- Speciation occurs without gene flow

... However, many species hybridize and remain distinct despite introgression, even with extreme gene flow (e.g., *Anopheles* mosquitoes^[1], *Heliconius* butterflies ^[2,3])

^[1] Fontaine et al. 2015; ^[2] Martin et al. 2019;

^[3] Rosser, Seixas et al. 2024

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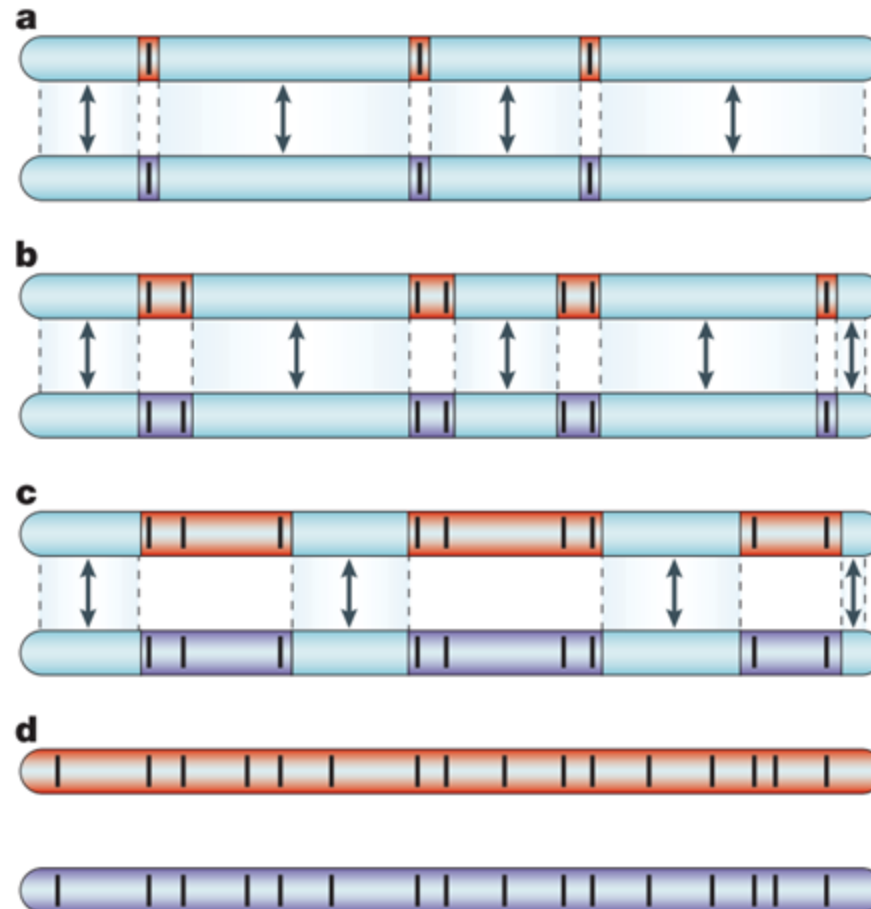
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^[1] Fontaine et al. 2015; ^[2] Martin et al. 2019;
^[3] Rosser, Seixas et al. 2024

Genic view of Speciation

Wu (2001)

- Gene as the unit of speciation
- Speciation can occur with or without gene flow
- ‘Speciation genes’ (aka barrier loci) reduce gene flow locally in the genome (which continues in other genomic regions)



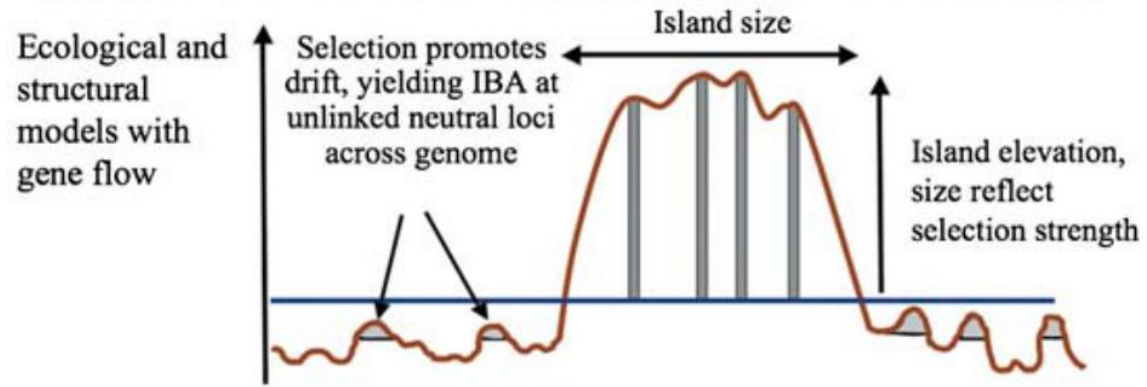
Divergent loci resist
gene flow

Gene flow continues but
linkage builds and
divergent regions grow

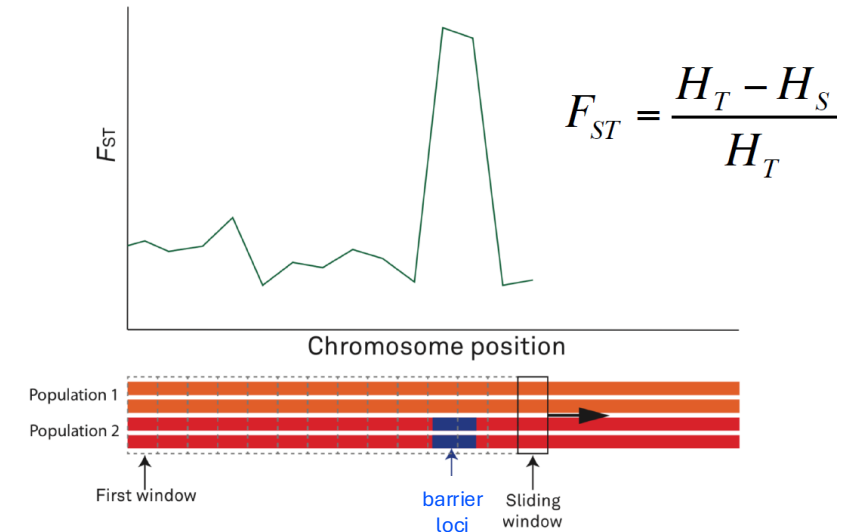
Complete reproductive
isolation evolves

Genomic Islands of Speciation

Turner et al (2005) PloS Biology; Turner & Hahn (2007) MBE;

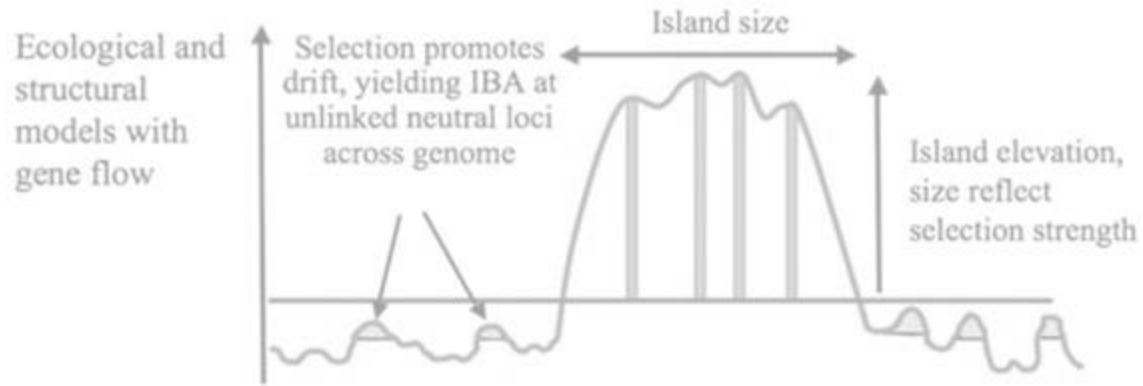


Nosil et al. (2012) Mol Ecol

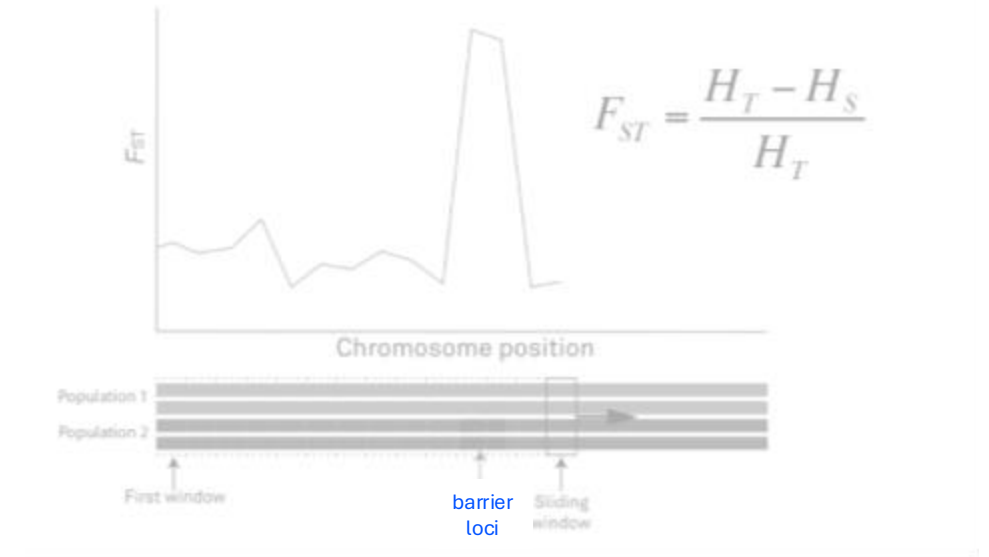


Genomic Islands of Speciation

Turner et al (2005) PloS Biology; Turner & Hahn (2007) MBE;



Nosil et al. (2012) Mol Ecol



Islands of Speciation or Mirages?

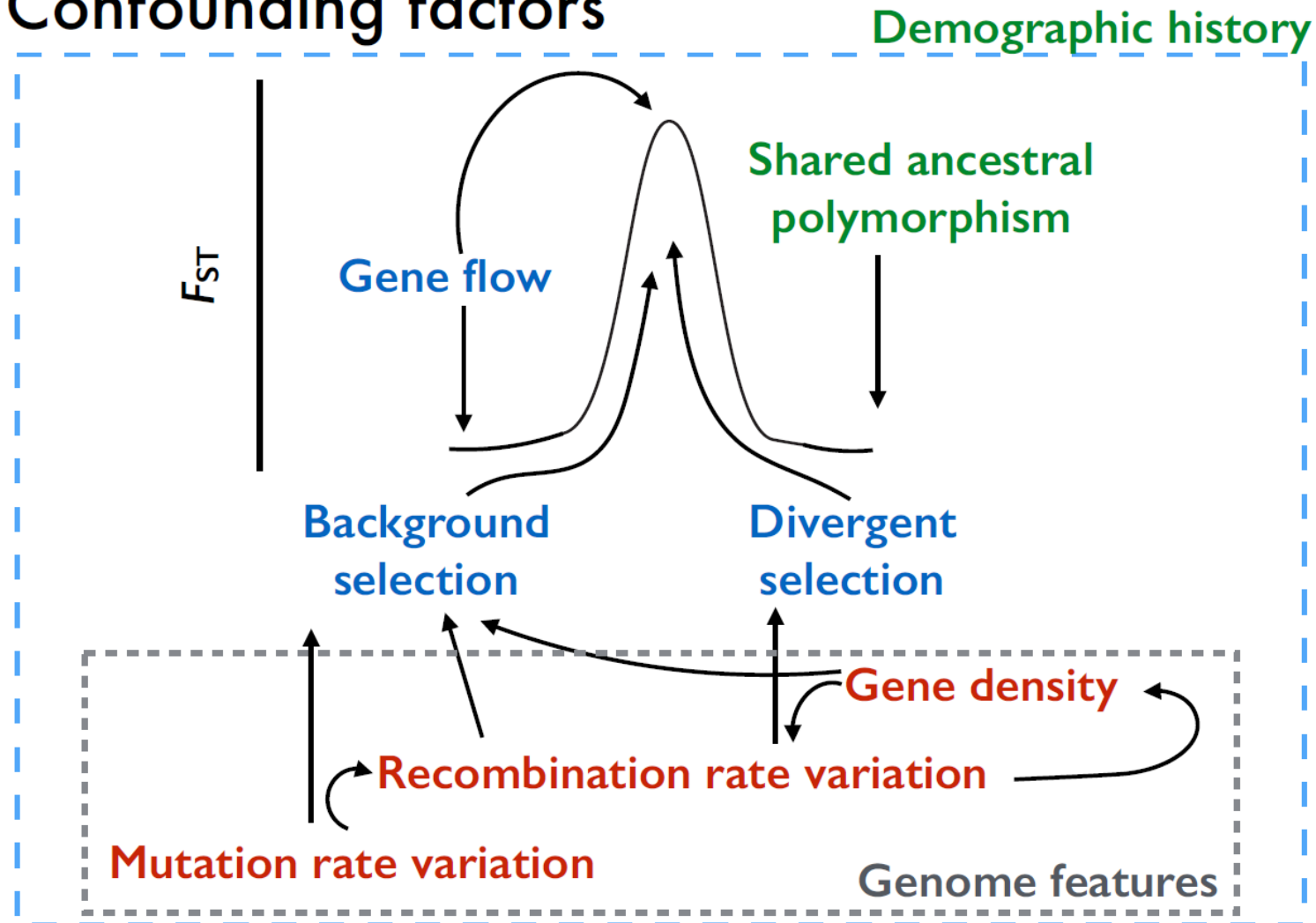
Noor & Bennett (2009) Heredity

Cruickshank & Hahn (2014) Mol Ecol.

Alternative explanations:

- Background selection or hitchhiking
- Local adaptation in isolation
- Rearrangement predating species split
- Shared ancestral polymorphism

Confounding factors



Complementing Genome Scans

Measure genomic factors that contribute to the differentiation landscape

- Gene density (genome annotation)
- Recombination rate (e.g. from crosses)

Sampling strategies

- Systems of parallel divergence or speciation
- Allopatric vs sympatric comparisons

Independent evidence for barrier loci

- test for a link between the genotype and the phenotype (e.g. QTL mapping), and separately test for selection on the phenotype

Making sense of genomic islands of differentiation in light of speciation

Jochen B. W. Wolf^{1,2} and Hans Ellegren¹



Interpreting differentiation landscapes in the light of long-term linked selection

Reto Burri^{1,2}

COMMENTARY

Parallelism in genomic landscapes of differentiation, conserved genomic features and the role of linked selection

H. ELLEGREN*  & J. B. W. WOLF*†

RESPONSE

Land ahoy? Navigating the genomic landscape of speciation while avoiding shipwreck

A. M. WESTRAM*  & M. RAVINET† 

TARGET REVIEW

Interpreting the genomic landscape of speciation: a road map for finding barriers to gene flow

M. RAVINET*† , R. FARIA‡§¶, R. K. BUTLIN¶**, J. GALINDO††, N. BIERNE‡‡, M. RAFAJLOVIĆ§§ , M. A. F. NOOR¶¶ , B. MEHLIG§§ & A. M. WESTRAM¶



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Interpreting the genomic landscape of introgression

Simon H Martin and Chris D Jiggins

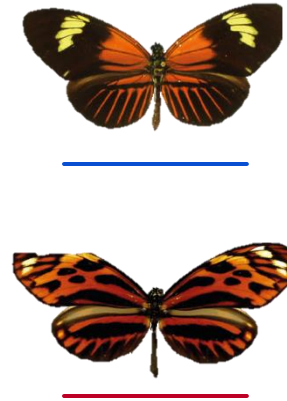
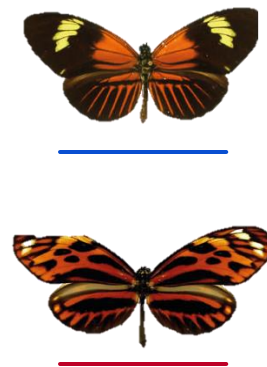
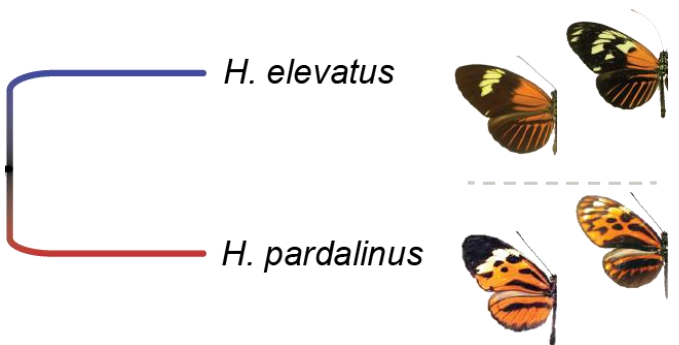


Some real-world examples

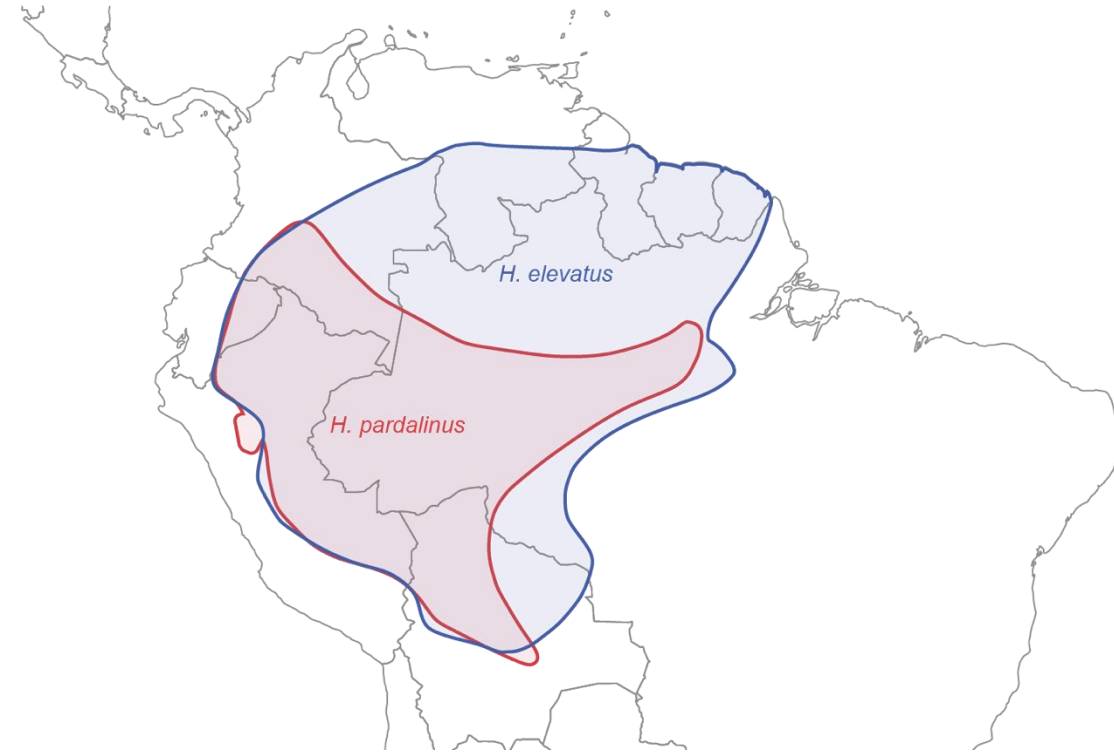
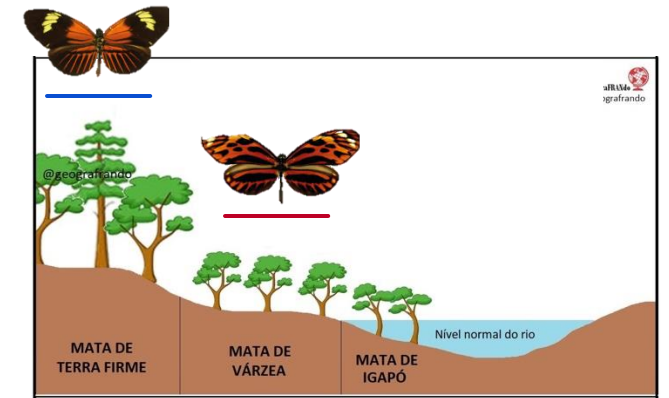
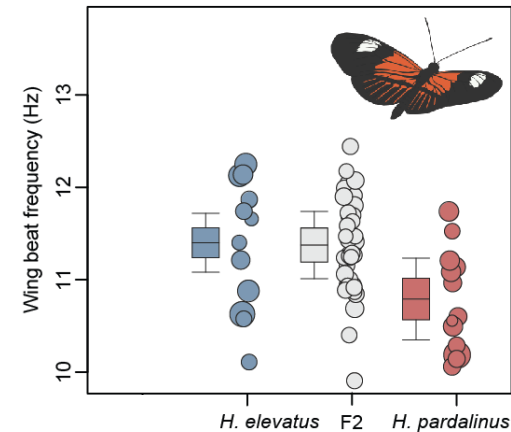
Hybrid speciation driven by multilocus introgression of ecological traits

Neil Rosser^{1,2,20}✉, Fernando Seixas^{1,20}, Lucie M. Queste², Bruna Cama², Ronald Mori-Pezo^{3,4}, Dmytro Kryvokhyzha^{1,5}, Michaela Nelson², Rachel Waite-Hudson², Matt Goringe², Mauro Costa⁶, Marianne Elias^{7,8}, Clarisse Mendes Eleres de Figueiredo^{9,10}, André Victor Lucci Freitas¹¹, Mathieu Joron¹², Krzysztof Kozak⁸, Gerardo Lamas¹³, Ananda R. P. Martins¹⁴, W. Owen McMillan⁸, Jonathan Ready^{9,10}, Nicol Rueda-Muñoz¹⁵, Camilo Salazar¹⁵, Patricio Salazar¹⁶, Stefan Schulz¹⁷, Leila T. Shirai¹¹, Karina L. Silva-Brandão¹⁸, James Mallet^{1,20} & Kanchon K. Dasmahapatra^{2,19}

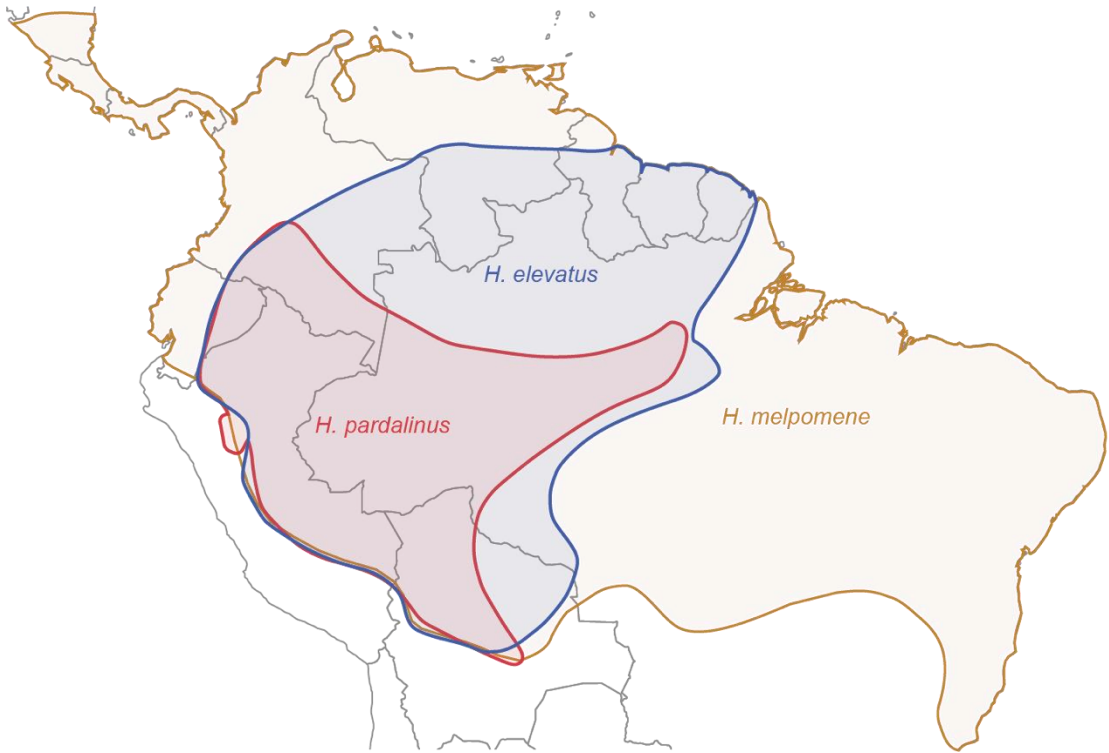
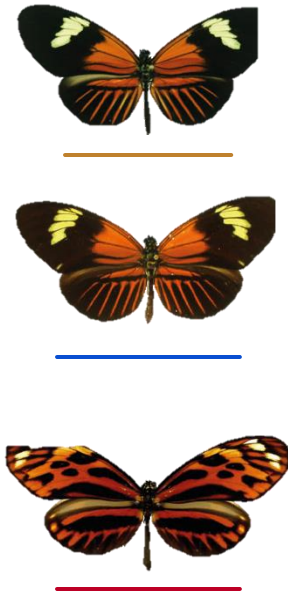
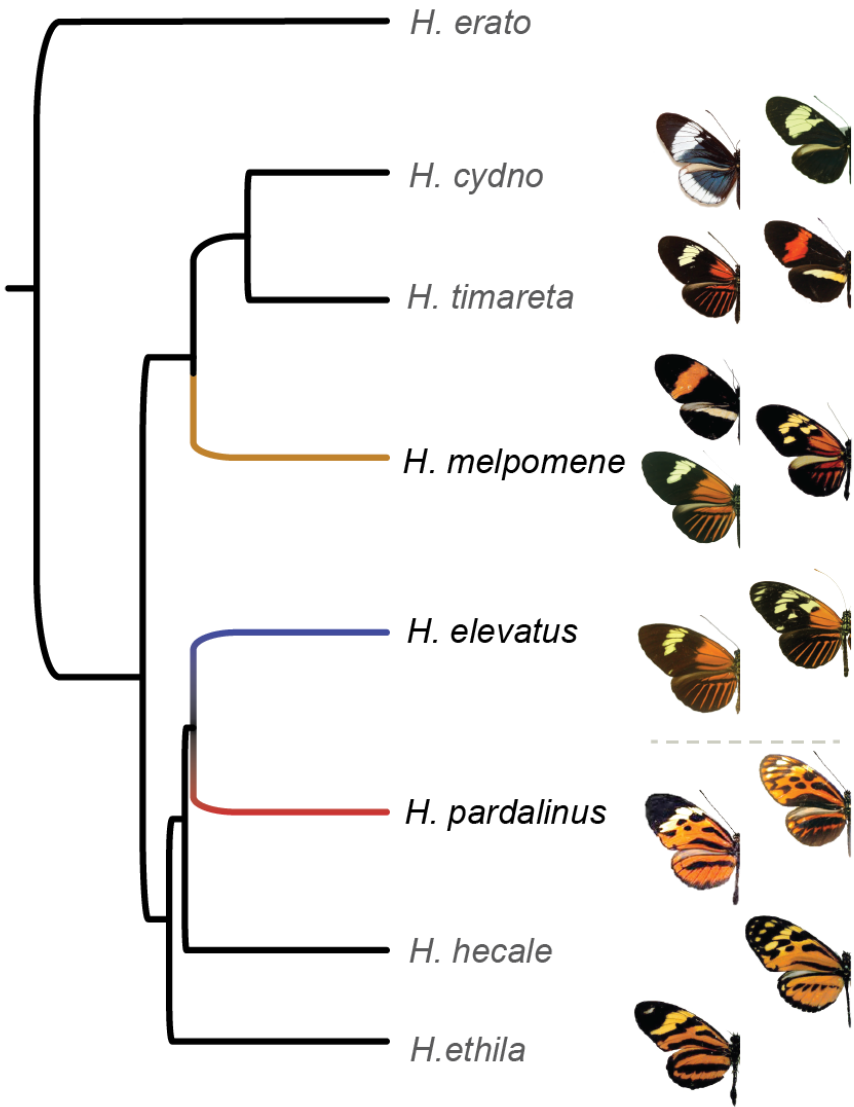
- Genetically close
- Highly sympatric
- Phenotypically distinct
- Different male pheromone bouquets
- Strong assortative mating
- Distinct ecologies (habitat and host-plant)



flight dynamics



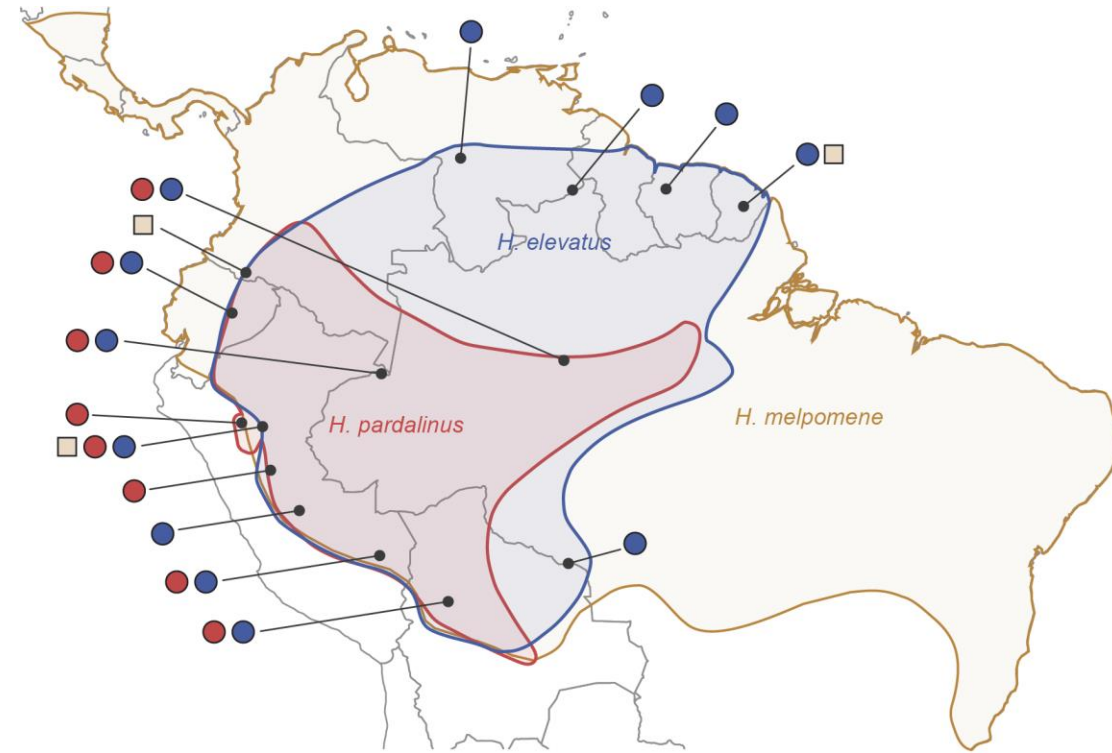
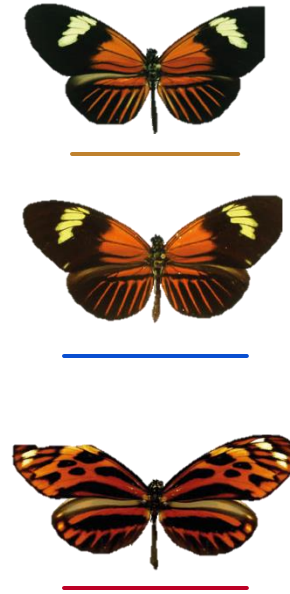
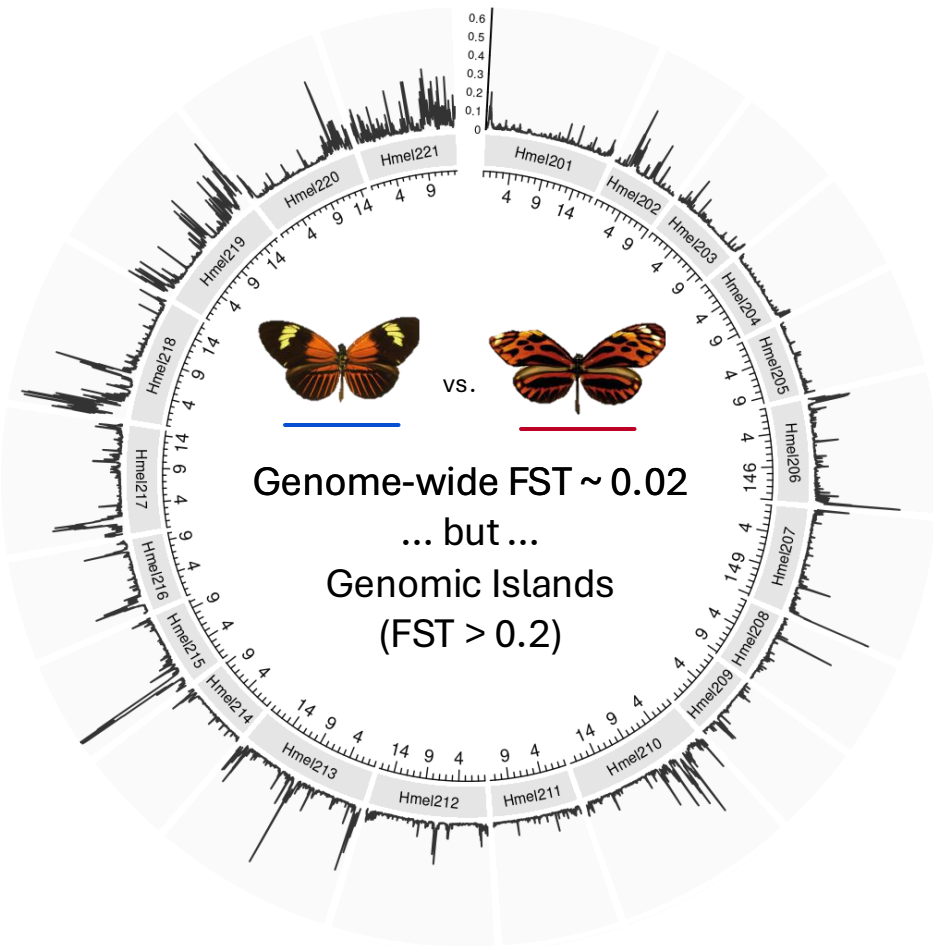
Hybrid speciation driven by multilocus introgression of ecological traits



Hybrid speciation driven by multilocus introgression of ecological traits

Sampling & Sequencing

- WGS sequencing
- 14 locations: sympatry and allopatric



Hybrid speciation driven by multilocus introgression of ecological traits

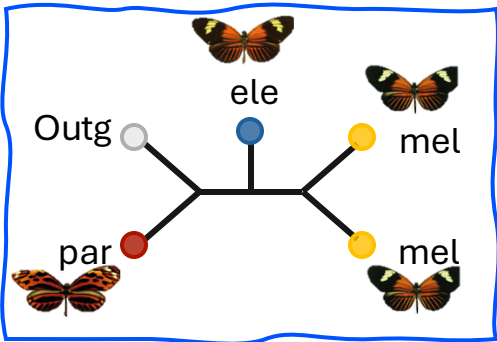
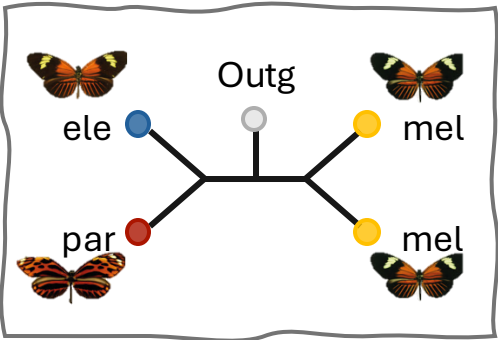
QTL mapping of species-specific traits

-  Colour pattern FW
-  Colour pattern HW
-  Wing shape FW
-  Wing shape HW
-  Host plant
-  Flight
-  Male preference
-  Pheromones

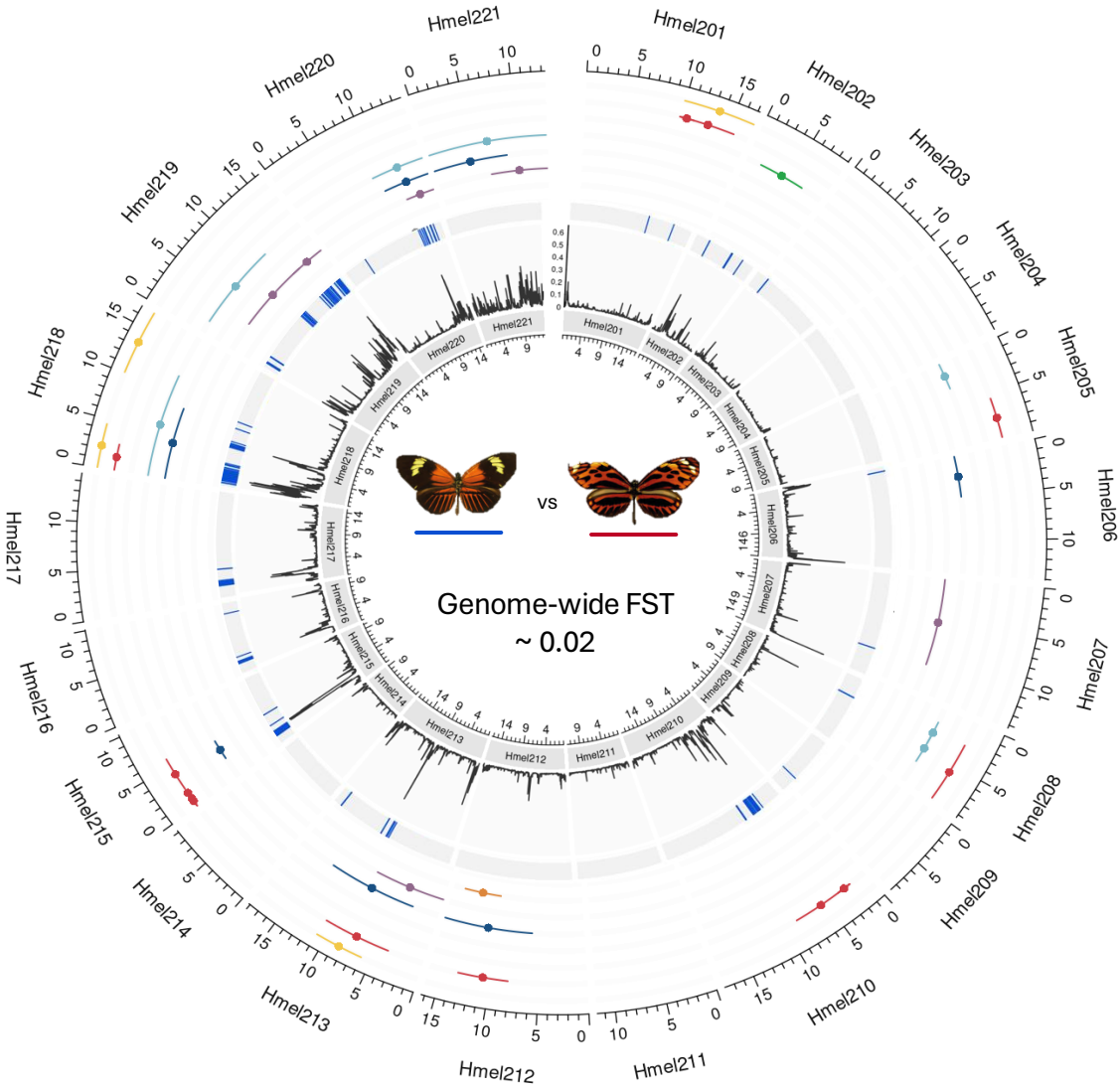


Neil Rosser

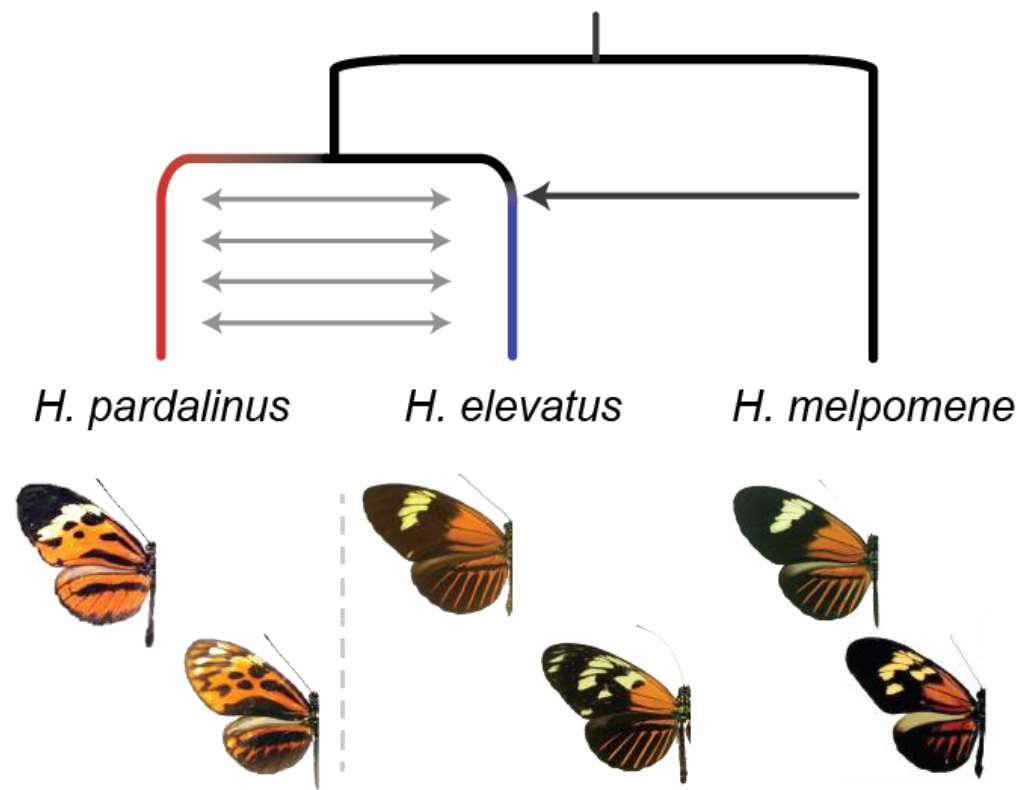
Genome Scans of introgression (TWISST)



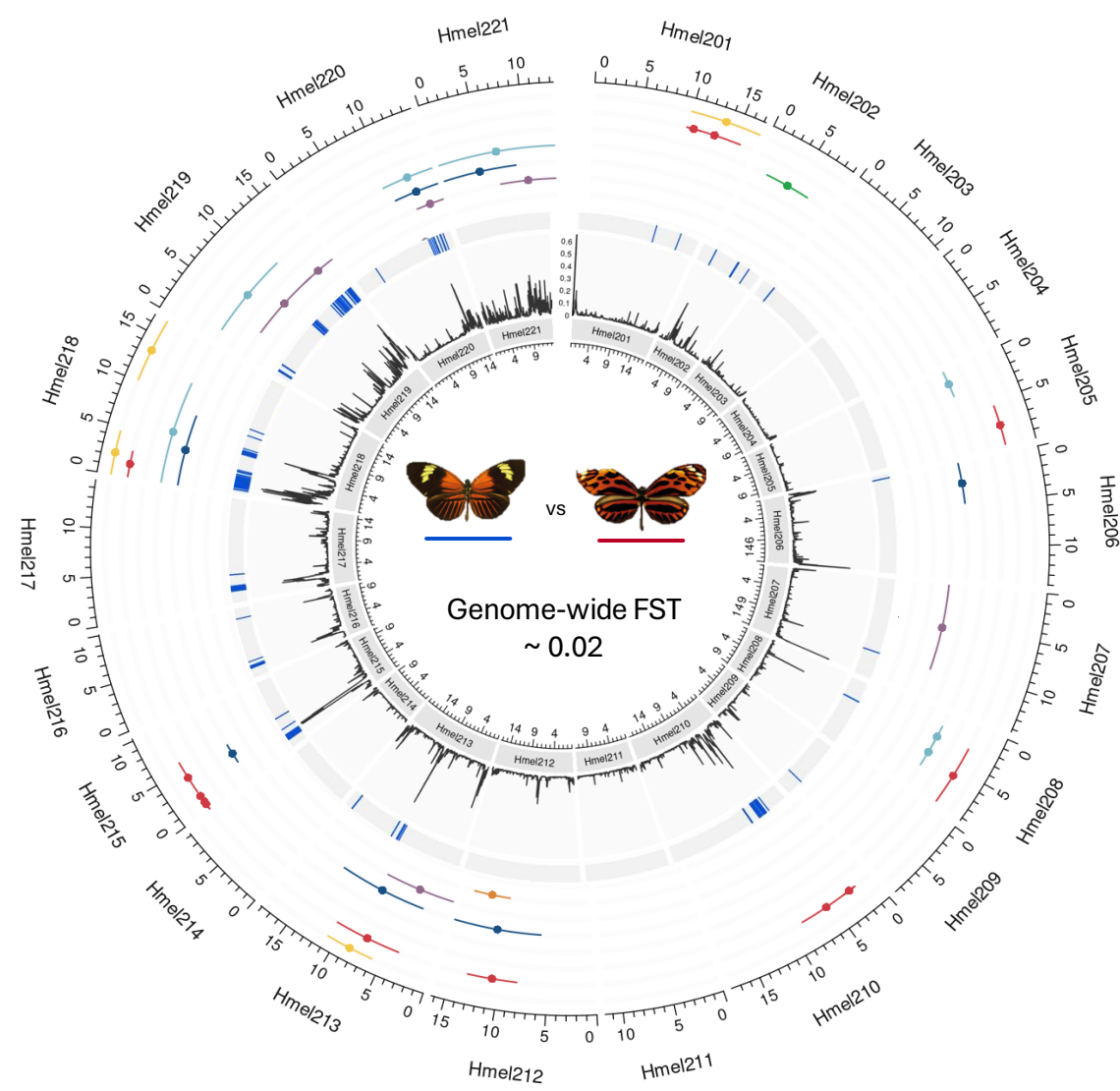
F_{ST} peaks $\approx$ QTL regions $\approx$ melp. introgression



Hybrid speciation driven by multilocus introgression of ecological traits

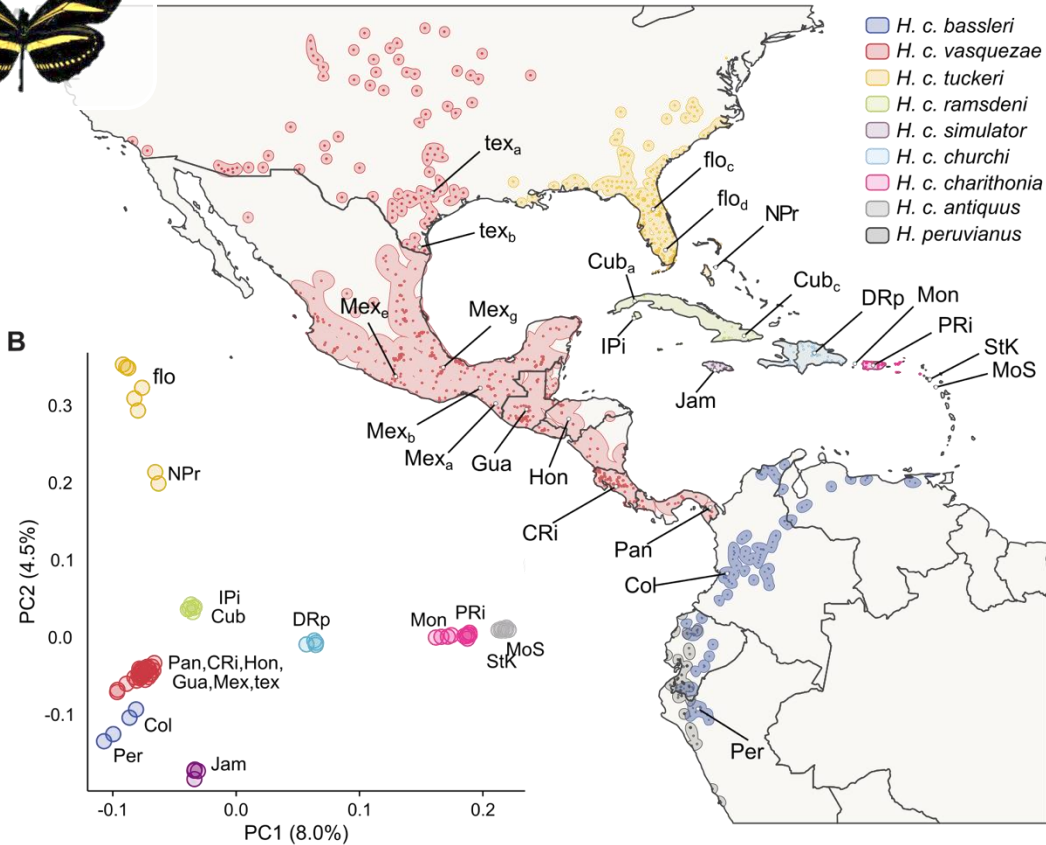


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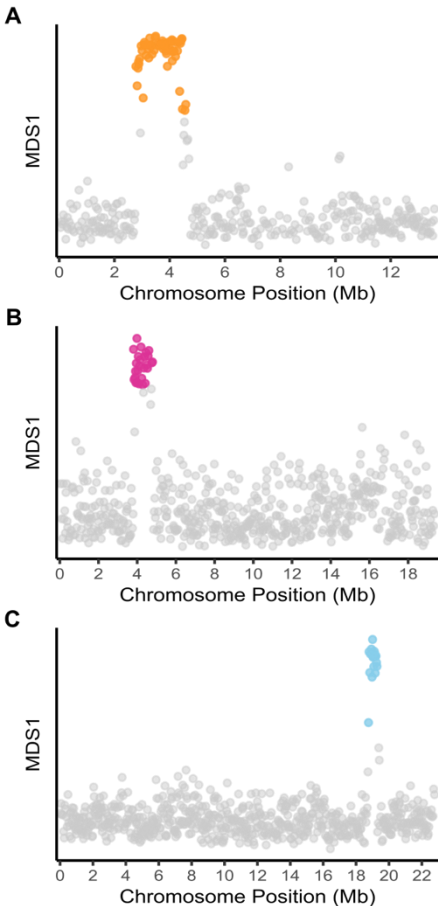


Ancient inversion polymorphisms are locally adaptive in a widespread butterfly species

Fernando Seixas¹, Sarah Dendy¹, Shuzhe Guan¹, Neil Rosser¹, Nick Grishin^{2,3}, Neil Davies⁴, Lawrence E. Gilbert Jr.⁵, W. Owen McMillan⁶, James Mallet¹

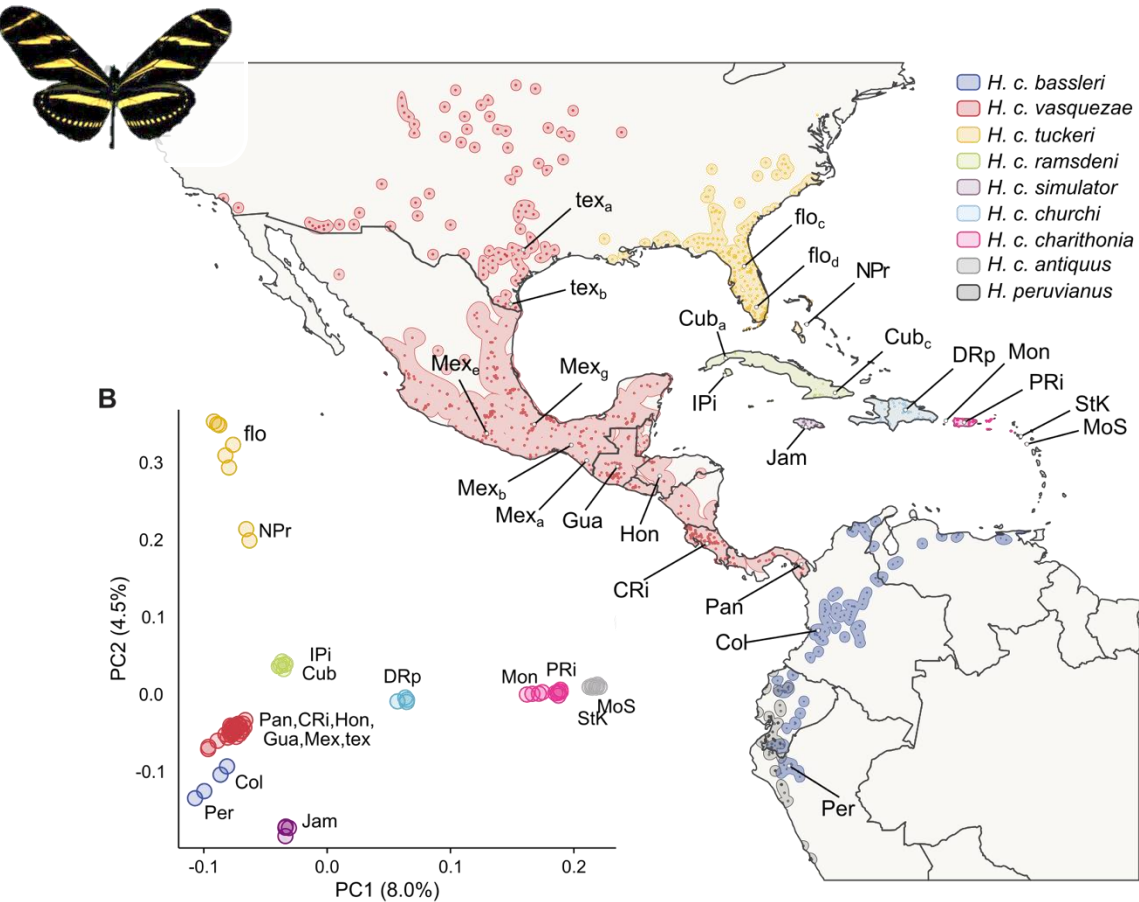


Local PCA (Li & Ralph 2019 Genetics)

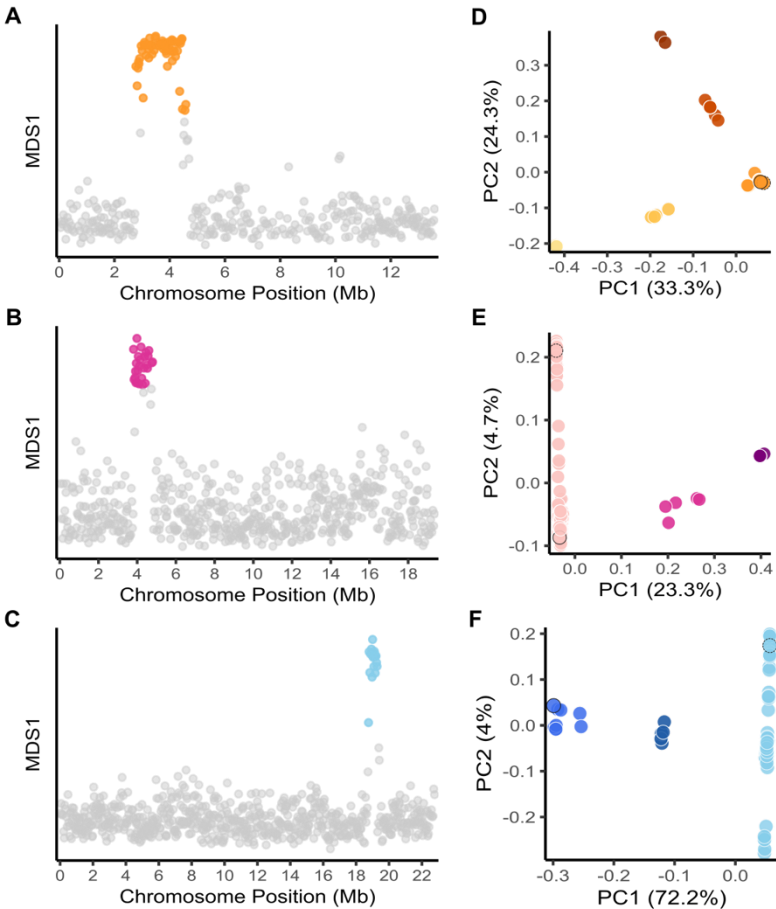


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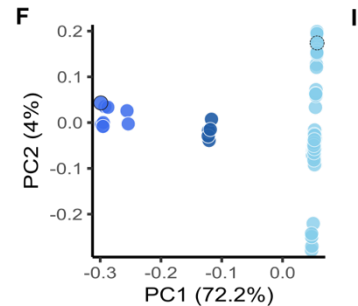
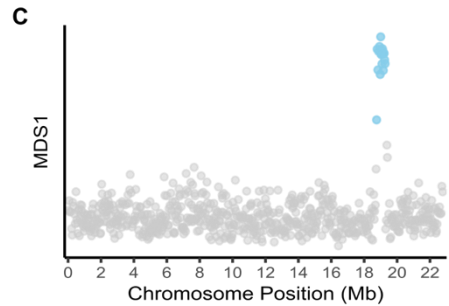
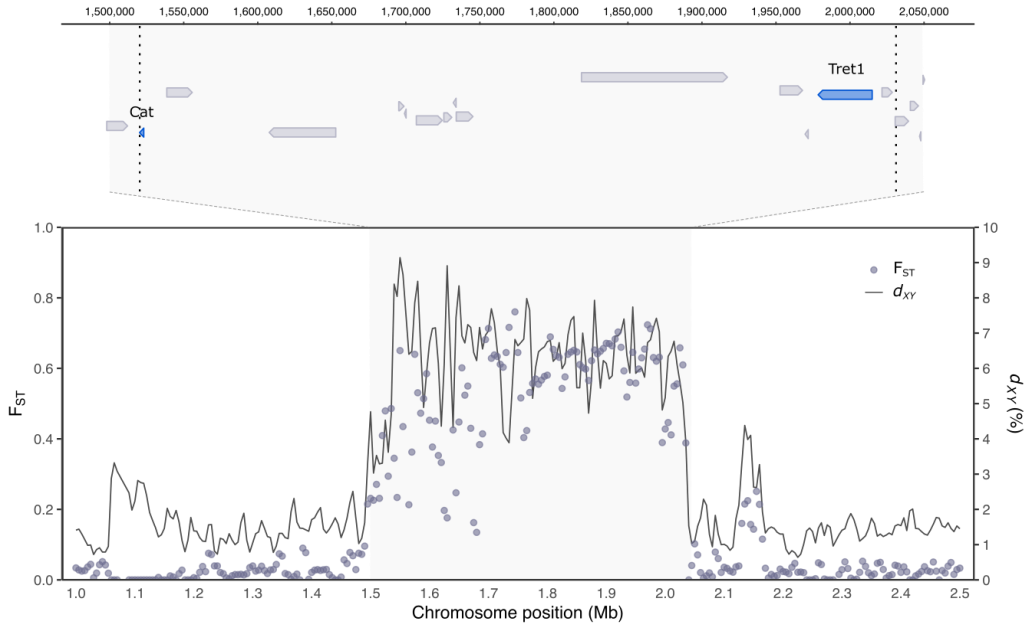
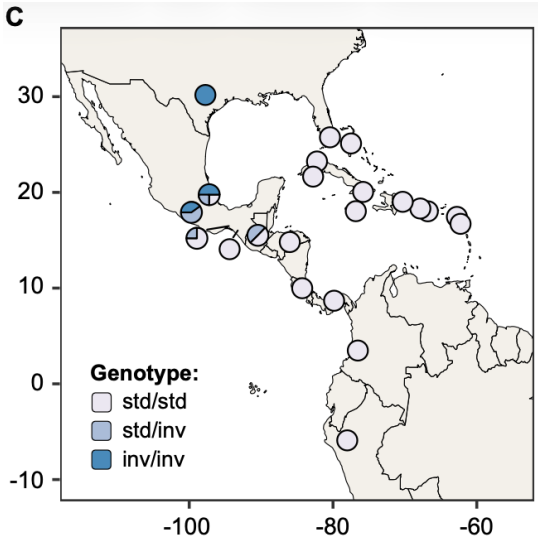
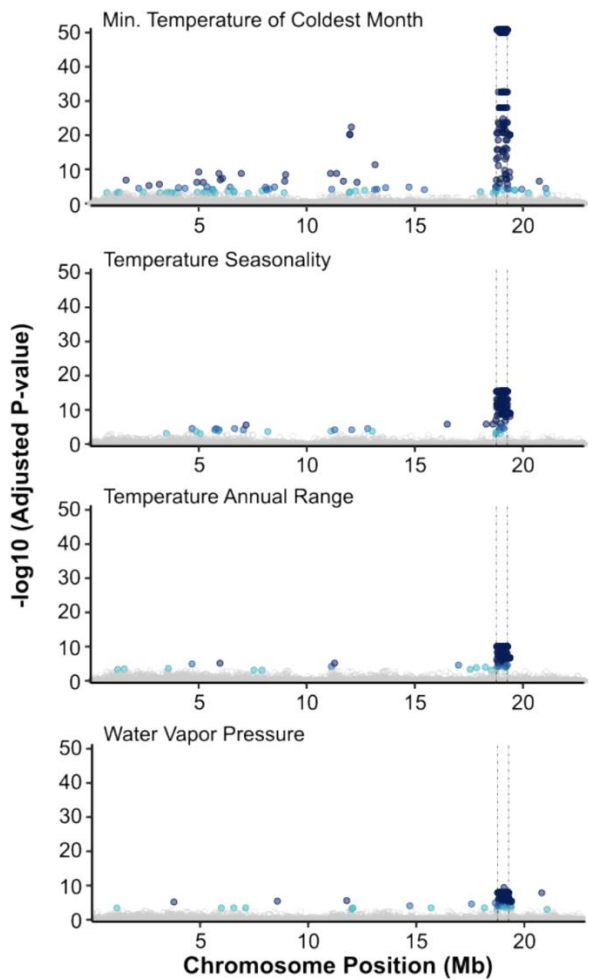


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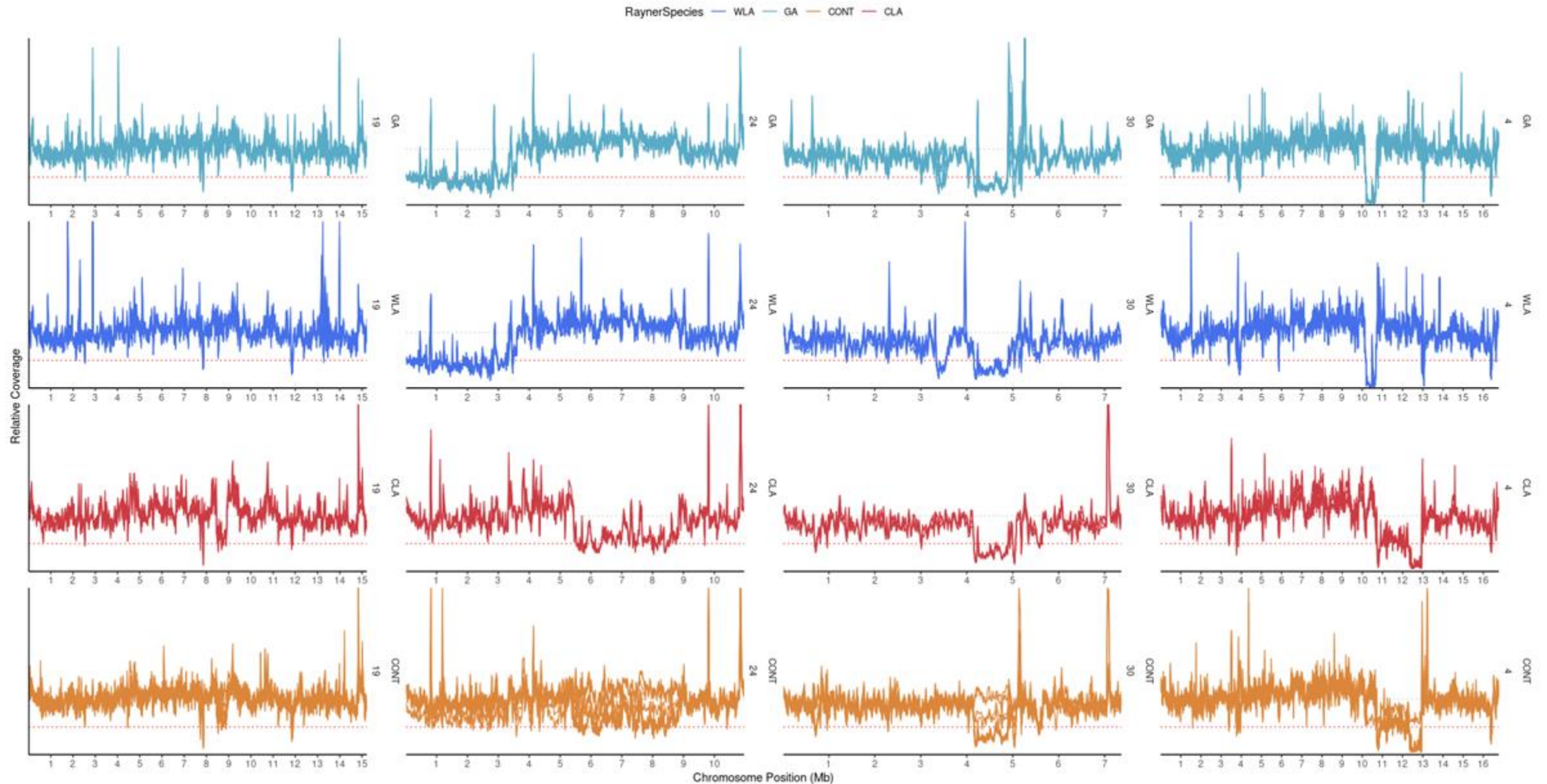


Ancient inversion polymorphisms are locally adaptive in a widespread butterfly species

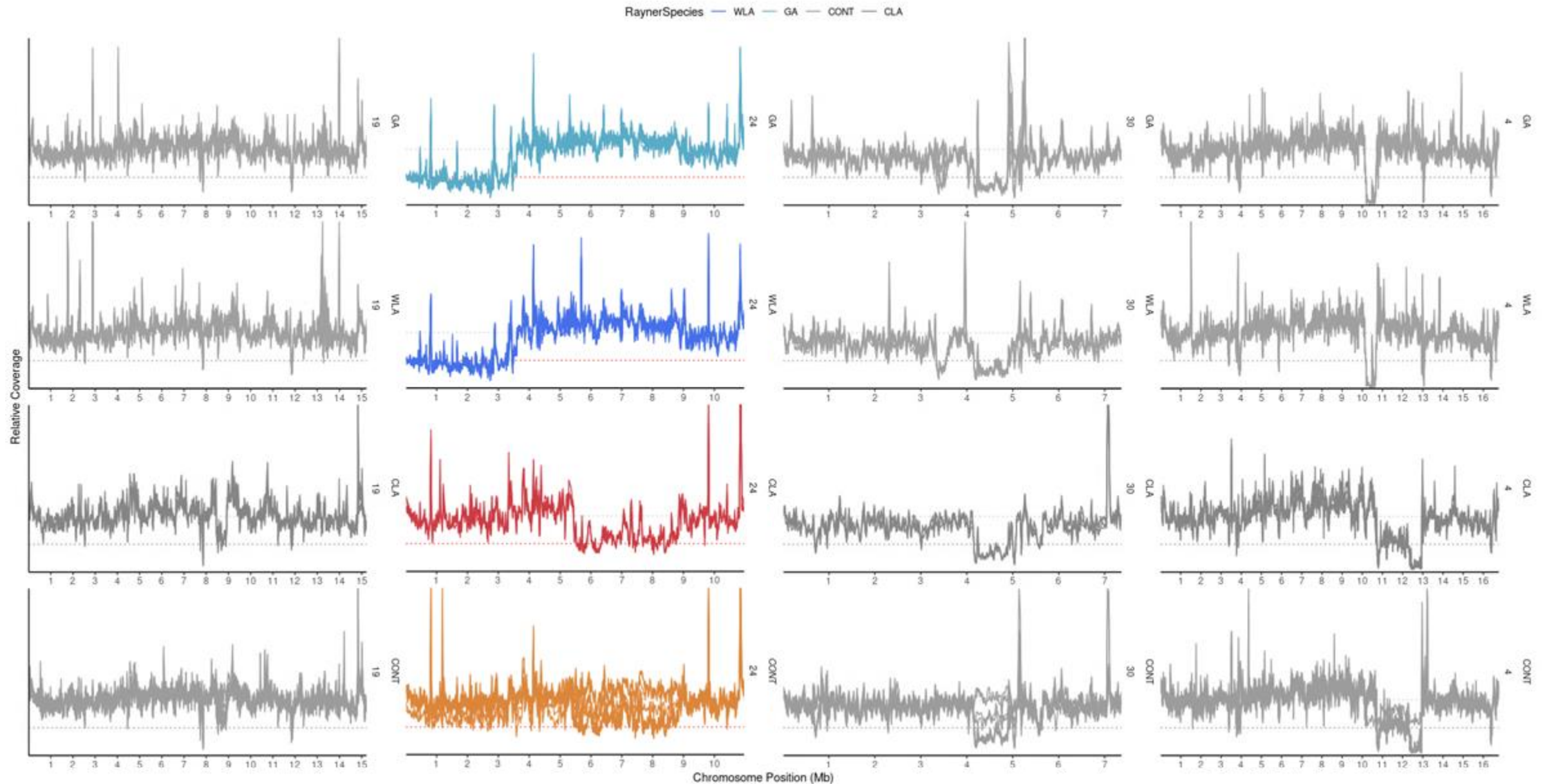
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Genomic Scans for depth of coverage reveal highly polymorphic regions



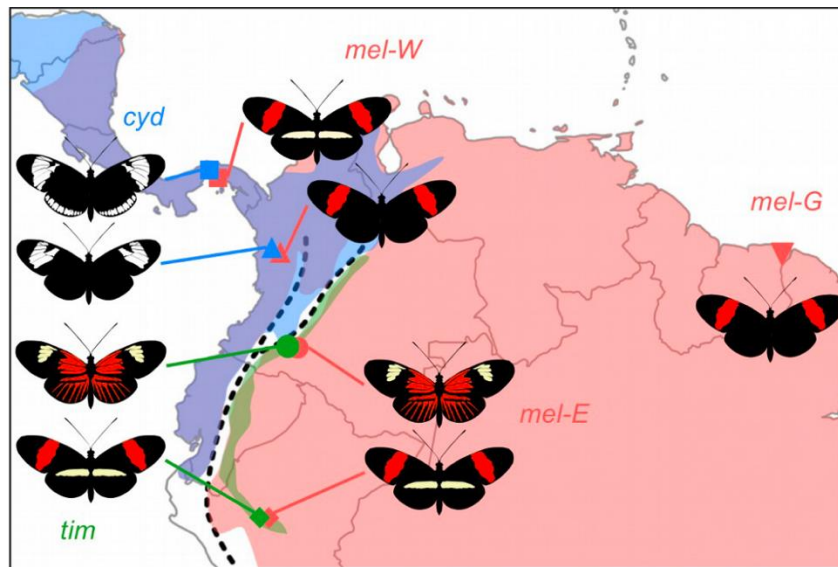
Genomic Scans for depth of coverage reveal highly polymorphic regions



Exercises

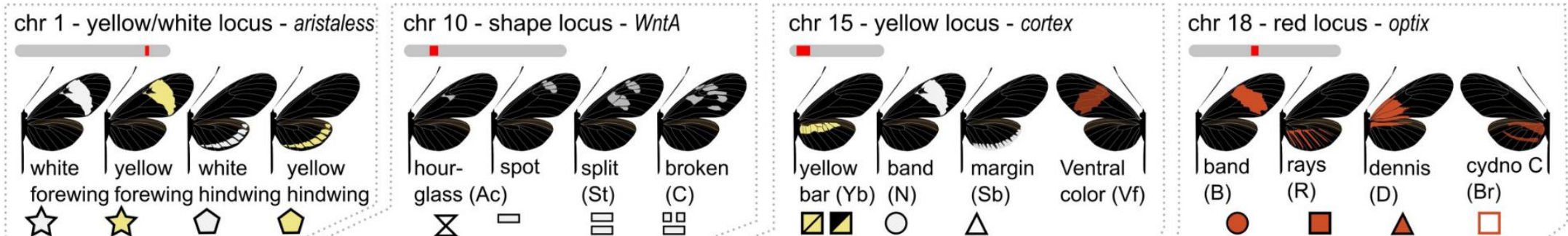
We will now apply genome scans to our *Heliconius* dataset (Martin et al 2019)

- Divergence (d_{XY}) and differentiation (F_{ST}) between *H. melpomene* populations.
- Introgression (f_d) between *H. melpomene* and *H. timareta* or *H. cydno*



mel-W | ■ *H. melpomene rosina*
▲ *H. m. vulcanus*
mel-E | ● *H. m. malleti*
◆ *H. m. amaryllis*
mel-G | ▼ *H. m. melpomene*

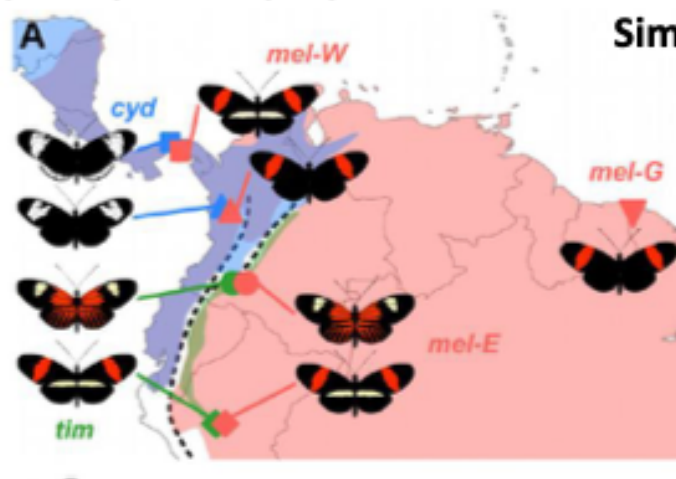
cyd | ■ *H. cydno chioneus*
▲ *H. c. zelinde*
tim | ● *H. timareta florencia*
◆ *H. t. thelxinoe*



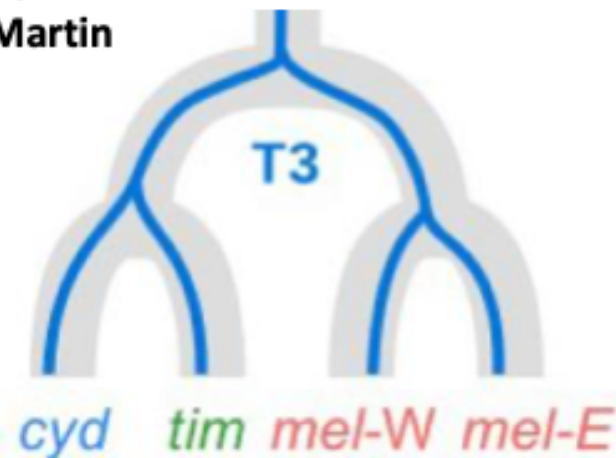
Supplementary Slides for those interested

Detecting regions under divergent selection and barriers to gene flow

- If high gene flow $\rightarrow F_{ST}$ is a good measure for detecting regions under divergent selection or barrier loci
- If the taxa are divergent enough $\rightarrow d_{xy}$ may work best, particularly if levels of gene flow are not very high and in cases of secondary contact. Ideally correct for differences in π with an outgroup.
- If the taxa are very young and gene flow is not very high, f_d or TWISST might help if allopatric and parapatric populations were sequenced



Simon Martin



Detecting regions under divergent selection and barriers to gene flow

- If rates of gene flow between the two taxa compared is high, F_{ST} is a good measure for detecting regions under divergent selection or barrier loci
- If the taxa are divergent enough, d_{xy} may work best, particularly if levels of gene flow are not very high and in cases of secondary contact. Ideally correct for differences in π with an outgroup.
- If the taxa are very young and gene flow is not very high, f_d or TWISST might help if allopatric and parapatric populations were sequenced
- If there is no gene flow, it is better to search for signatures of selective sweeps (e.g. iHS, XP-EHH, Tajima's D). However, inferring if these regions are involved in speciation is difficult.

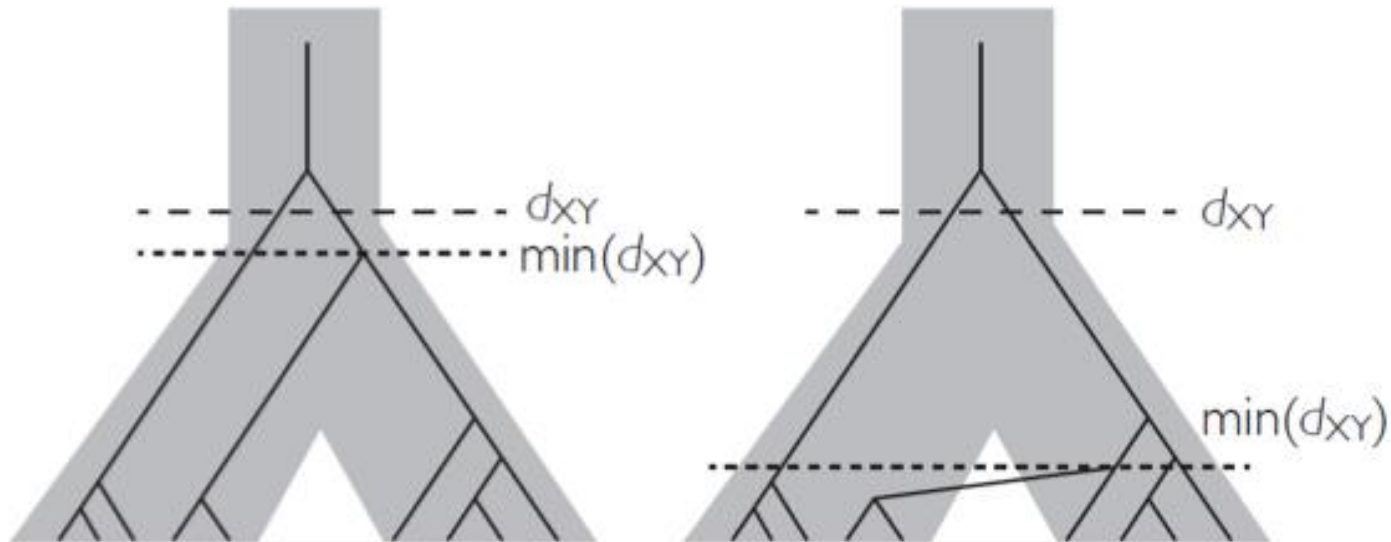
Absolute measures of divergence

$$d_{XY} = \sum_{ij} x_i y_j d_{ij}$$

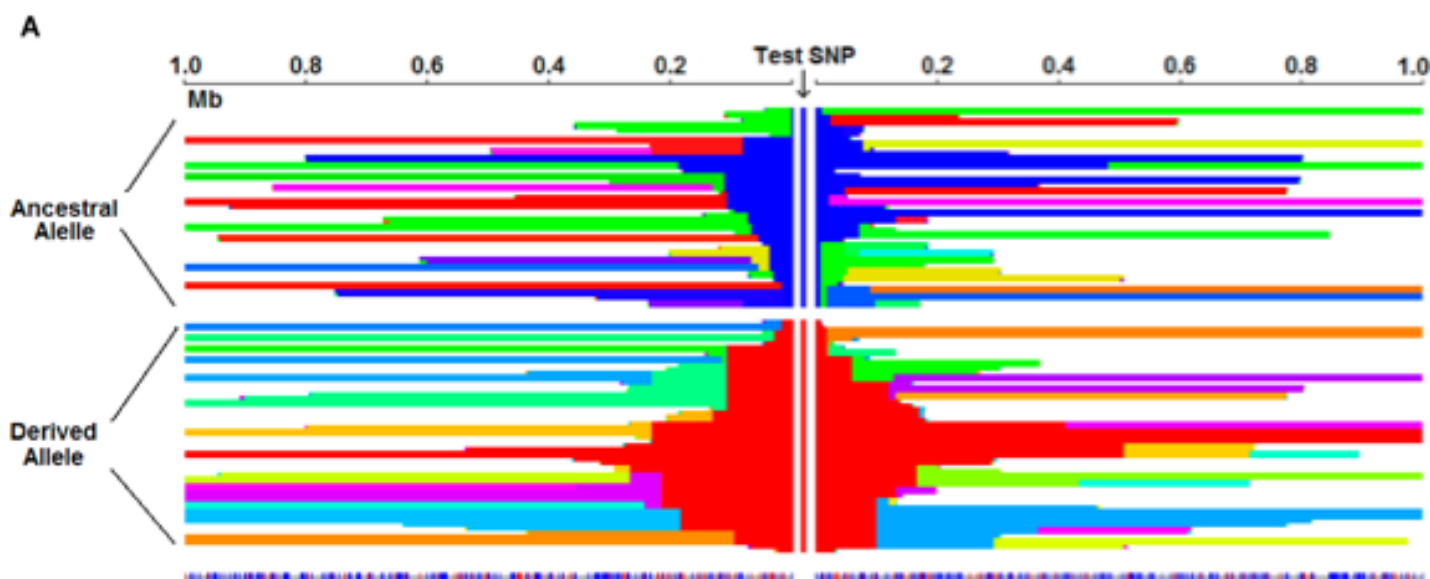
Average number of pairwise differences between two populations

Pop A	Pop B
ACTGTC	ATTAGC
ATTGTC	ACTGGC
ACTGTC	ACTAGC
ATTGTC	ATTAGC

Here d_{XY} is
0.375



iHS and XP-EHH



iHS: within a population

iHS (integrated haplotype score) compares haplotype lengths **within a population**

-> an allele under selection will lead to increased haplotype length relative to other haplotypes in the same region

-> useful to detect **ongoing/incomplete sweeps**

XP-EHH: between populations

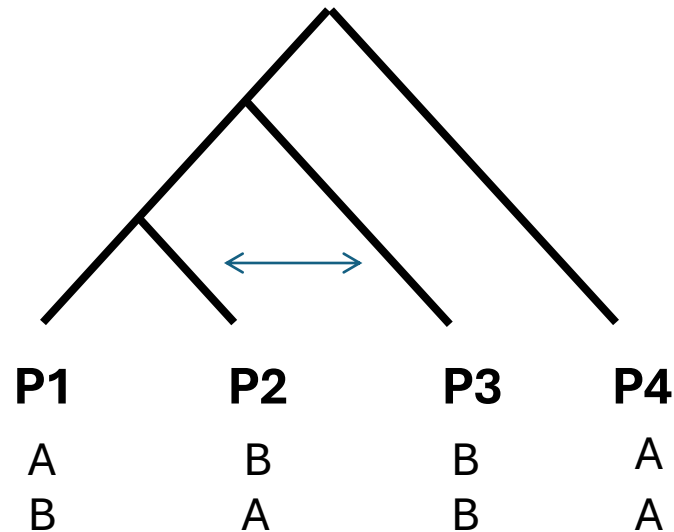
XP-EHH (cross population extended haplotype homozygosity) compares haplotype lengths **between populations**

-> a population that had a sweep has increased haplotype lengths relative to the haplotypes in the other population in the same region

-> most powerful with **complete sweeps** restricted to one population

Sliding window introgression: f_d

f_d can be applied to smaller number of ABBA and BABA sites than D and is thus ideal for sliding windows. ABBA and BABA patterns are computed from allele frequencies and the f test of the four populations is standardized by the maximum value it could get which would be the scenario of complete mixing between P2 and P3. P2 and P3 are thus both set to PD which is the taxon with higher derived allele frequency of P2 and P3.



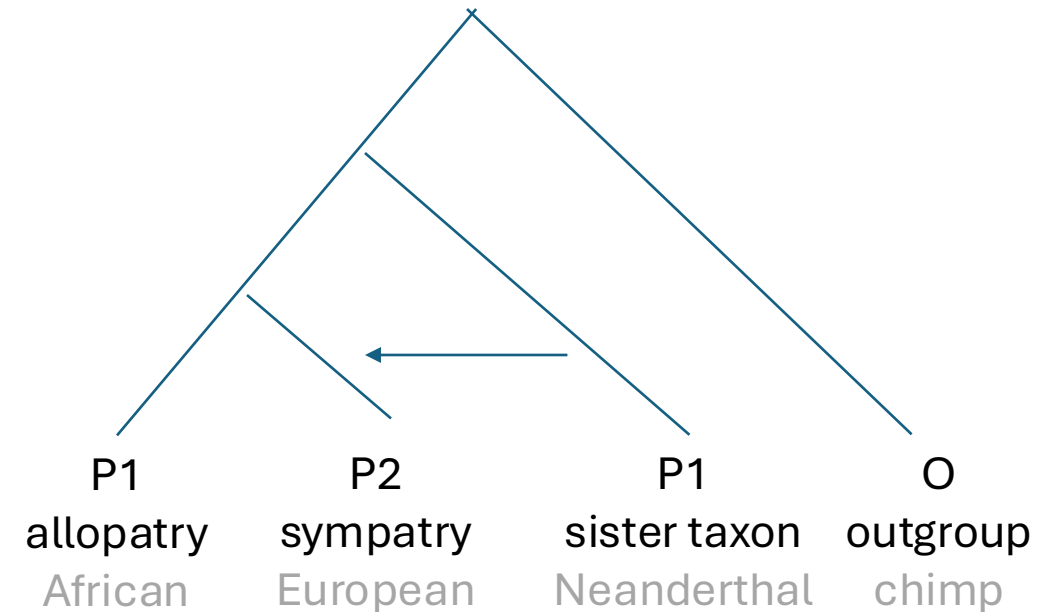
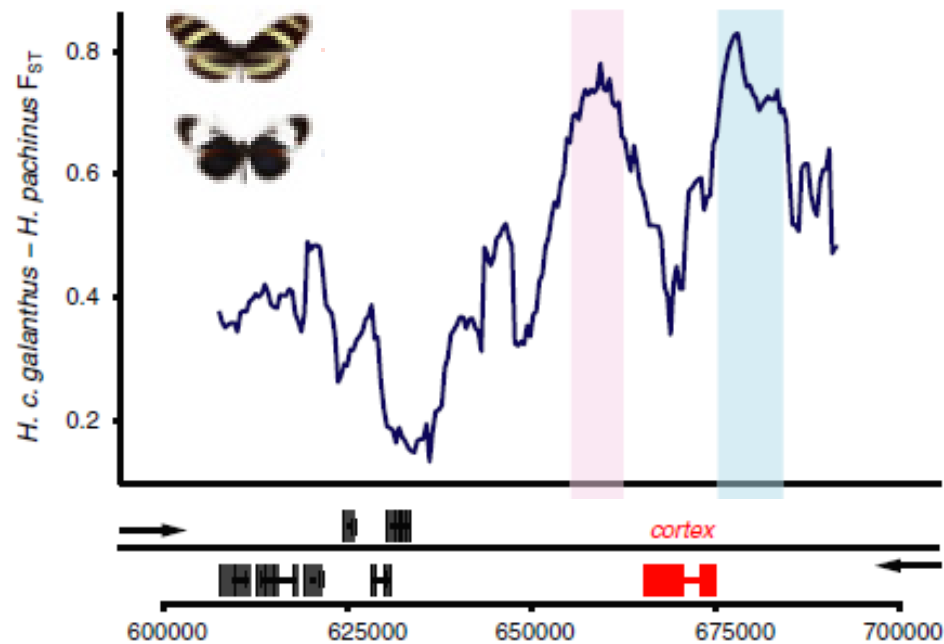
$$C_{ABBA}(i) = (1 - \hat{p}_{i1})\hat{p}_{i2}\hat{p}_{i3}(1 - \hat{p}_{i4})$$

$$C_{BABA}(i) = \hat{p}_{i1}(1 - \hat{p}_{i2})\hat{p}_{i3}(1 - \hat{p}_{i4})$$

$$\hat{f}_d = \frac{S(P_1, P_2, P_3, O)}{S(P_1, P_D, P_D, O)}$$

PD=P2 or P3
(taxon with higher
derived allele
frequency)

f_d can be used to find regions of reduced gene flow if allopatric and sympatric populations exist or alternatively, of adaptive introgression



TWISST: Visualizing gene trees across the genome

